



GATE ME 90 Days Challenge

Day 12/90 · Practice Set with Detailed Solutions

SUBJECT

Weekly Revision

TODAY'S TOPIC

Week 2 Revision

COUNTDOWN

98 days to GATE

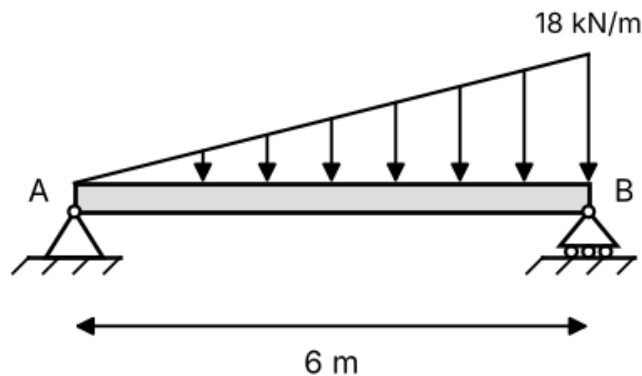
How to use this set: 15 questions (8 MCQ, 7 NAT) on all five topics of Week 2: SFD and BMD; bending and shear stresses; verbal ability; deflection of beams; torsion, thin cylinders and springs (3 questions each). Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.

Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 (MCQ)

The simply supported beam AB shown (span 6 m) carries a load that increases linearly from zero at A to 18 kN/m at B. The maximum bending moment and its location are



(A) 40.50 kN·m at mid-span

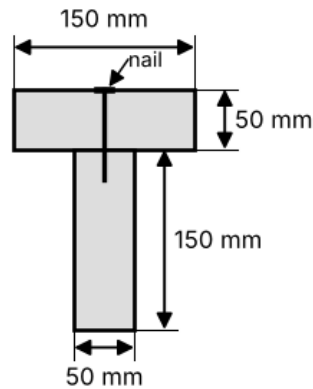
(B) 41.57 kN·m at 3.46 m from A

(C) 54.00 kN·m at 3.46 m from A

(D) 41.57 kN·m at 2.54 m from A

Q6 (NAT)

A T-beam is built by nailing a 150 mm × 50 mm flange plank on top of a 50 mm × 150 mm web plank, as shown. Each nail can safely carry 1.2 kN in shear. If the beam carries a vertical shear force of 3 kN, the maximum longitudinal spacing of the nails (in mm) is _____

**Q7** (MCQ)

Select the sentence that is grammatically correct.

(A) Neither the manager nor the engineers was present at the review.

(C) Every one of the bolts was checked before assembly.

(B) The report shows that each of the samples have failed.

(D) Scarcely had the test begun when the power fails.

Q8 (MCQ)

Select the word that is most nearly OPPOSITE in meaning to FRUGAL.

(A) thrifty

(C) extravagant

(B) prudent

(D) modest

Q9 (MCQ)

Select the most appropriate word to fill the blank: The results of the first trial were _____, so the experiment had to be repeated.

(A) inconclusive

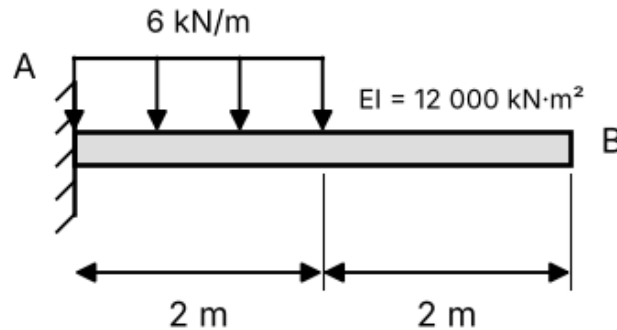
(C) unanimous

(B) conclusive

(D) definitive

Q10 (NAT)

The cantilever AB shown (fixed at A, length 4 m, $EI = 12\,000\text{ kN}\cdot\text{m}^2$) carries a UDL of 6 kN/m only over the 2 m next to the fixed end. The deflection of the end B (in mm, round off to two decimal places) is _____

**Q11** (MCQ)

A simply supported beam of span L (constant EI) carries a point load P at a distance $L/3$ from the left support. The deflection under the load is

- (A) $PL^3/(48EI)$ (B) $8PL^3/(243EI)$
 (C) $4PL^3/(729EI)$ (D) $4PL^3/(243EI)$

Q12 (NAT)

A beam of span 6 m, fixed at both ends ($EI = 24\,000\text{ kN}\cdot\text{m}^2$), carries a point load of 40 kN at mid-span. The maximum deflection (in mm) is _____

Q13 (NAT)

A closed thin cylindrical vessel has internal diameter 1.2 m, length 3 m and wall thickness 12 mm. It is subjected to an internal pressure of 1.5 MPa ($E = 200\text{ GPa}$, $\nu = 0.3$). The increase in its length (in mm) is _____

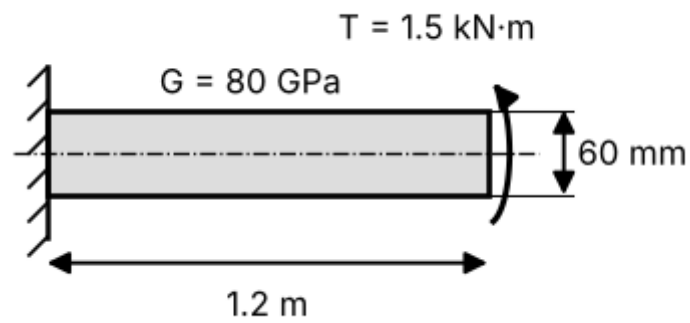
Q14 (MCQ)

Two close-coiled helical springs of the same material and the same unloaded length are placed one inside the other (concentric) and together carry an axial load of 1000 N. The outer spring has 10 mm wire, 80 mm mean coil diameter and 8 turns; the inner spring has 6 mm wire, 48 mm mean coil diameter and 12 turns. The load carried by the outer spring is

- (A) 285.7 N (B) 714.3 N
 (C) 500.0 N (D) 625.0 N

Q15 (NAT)

The solid shaft shown (diameter 60 mm, length 1.2 m, $G = 80$ GPa) is fixed at one end and a torque of 1.5 kN·m is applied gradually at the other. The strain energy stored in the shaft (in J) is _____



Answer Key

Q	Type	Answer
Q1	MCQ	(B) 41.57 kN·m at 3.46 m from A
Q2	NAT	25.88 (accept 25.6 to 26.1)
Q3	MCQ	(D) 2 and 1.6
Q4	NAT	65.86 (accept 65.4 to 66.3)
Q5	MCQ	(A) at mid-height, $h/2$ from the apex
Q6	NAT	56.67 (accept 56.1 to 57.2)
Q7	MCQ	(C) Every one of the bolts was checked before assembly.
Q8	MCQ	(C) extravagant
Q9	MCQ	(A) inconclusive
Q10	NAT	2.33 (accept 2.31 to 2.35)
Q11	MCQ	(D) $4PL^3/(243EI)$
Q12	NAT	1.88 (accept 1.86 to 1.89)
Q13	NAT	0.225 (accept 0.223 to 0.227)
Q14	MCQ	(B) 714.3 N
Q15	NAT	13.26 (accept 13.1 to 13.4)

Detailed Solutions

Q1 Answer: (B) 41.57 kN·m at 3.46 m from A

Concept: The maximum bending moment occurs where the shear force is zero. For a triangular load the shear varies as a parabola.

Total load = $\frac{1}{2} \times 18 \times 6 = 54$ kN acting at $2L/3 = 4$ m from A $\rightarrow R_A = 54 \times 2/6 = 18$ kN, $R_B = 36$ kN

Intensity at x: $w = 3x$ kN/m $\rightarrow V(x) = 18 - 1.5x^2 = 0 \rightarrow x = \sqrt{12} = 3.46$ m

$M(x) = 18x - 0.5x^3 \rightarrow M_{\max} = 18(3.464) - 0.5(3.464)^3 = 62.35 - 20.78 = \mathbf{41.57 \text{ kN}\cdot\text{m at 3.46 m from A}}$ ($= wL^2/(9\sqrt{3})$)

Exam tip: The peak is NOT at mid-span (M there is 40.50 kN·m) and it lies towards the heavier end, at $L/\sqrt{3} = 0.577L$ from the zero-load end.

Q2 Answer: 25.88 (accept 25.6 to 26.1)

Concept: Find the reactions, then locate the zero-shear point between the supports; the hogging peak is at the support next to the overhang.

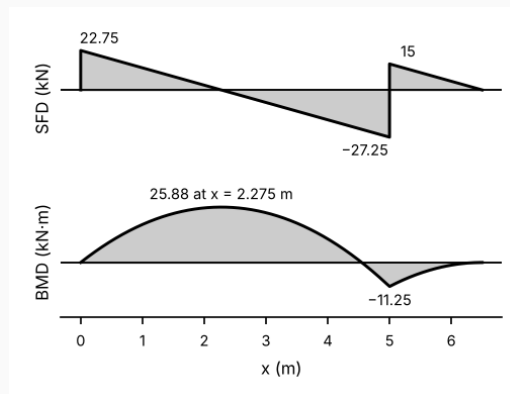
Total load = $10 \times 6.5 = 65$ kN at 3.25 m from A. Moments about A: $R_B \times 5 = 65 \times 3.25 \rightarrow R_B = 42.25$ kN; $R_A = 22.75$ kN

Zero shear: $22.75 - 10x = 0 \rightarrow x = 2.275$ m

$M_{\max} = R_A^2/(2w) = 22.75^2/20 = \mathbf{25.88 \text{ kN}\cdot\text{m}}$ (sagging)

Hogging at B = $10 \times 1.5^2/2 = 11.25$ kN·m. The SFD jumps by $R_B = 42.25$ kN at B (from -27.25 to $+15$ kN).

Exam tip: $R_A^2/(2w)$ is a quick shortcut whenever the zero-shear point lies inside a UDL that starts at the support.

**Q3 Answer: (D) 2 and 1.6**

Concept: Compare the standard results for a UDL and a central point load with the same total W .

UDL: $M_{\max} = WL/8$, $\delta_{\max} = 5WL^3/(384EI)$

Central load: $M_{\max} = WL/4$, $\delta_{\max} = WL^3/(48EI)$

Ratios: $(WL/4)/(WL/8) = 2$ and $(1/48)/(5/384) = 384/240 = 1.6 \rightarrow \mathbf{2 \text{ and } 1.6}$

Exam tip: Spreading a load always helps, but it helps the moment ($\times 1/2$) more than the deflection ($\times 0.625$).

Q4 Answer: 65.86 (accept 65.4 to 66.3)

Concept: Locate the neutral axis (centroid), find I about it, then $\sigma = My/I$ at the extreme fibre on the compression side (top, for sagging).

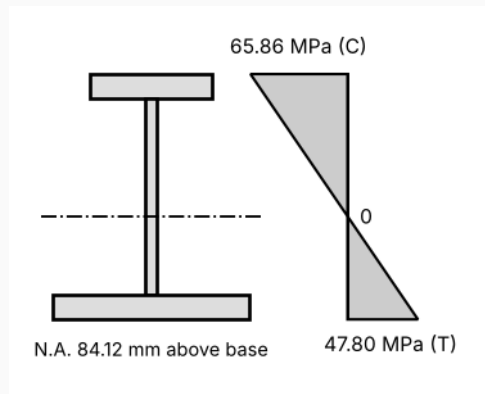
Areas: 3200 mm^2 at 10 mm , 1600 mm^2 at 100 mm , 2000 mm^2 at 190 mm (from the base). $\bar{y} = (32\,000 + 160\,000 + 380\,000)/6800 = 84.12 \text{ mm}$

$$I = \Sigma[bh^3/12 + A d^2] = 4.399 \times 10^7 \text{ mm}^4$$

$$M = PL/4 = 20 \times 5/4 = 25 \text{ kN}\cdot\text{m}; \text{ top fibre at } y = 200 - 84.12 = 115.88 \text{ mm}$$

$$\sigma_c = 25 \times 10^6 \times 115.88 / 4.399 \times 10^7 = \mathbf{65.86 \text{ MPa}}$$
 (bottom, tension: $25 \times 10^6 \times 84.12 / 4.399 \times 10^7 = 47.80 \text{ MPa}$)

Exam tip: In an unsymmetric section the larger stress is at the fibre farther from the N.A., which is on the side of the smaller flange.

**Q5 Answer: (A) at mid-height, $h/2$ from the apex**

Concept: $\tau = VQ/(Ib)$. For a triangle, both Q and the width b change with depth, so the peak need not be at the N.A.

At depth y from the apex: $b = By/h$, $Q = By^2(h - y)/(3h)$ and $I = Bh^3/36$, so $\tau = VQ/(Ib) = 12V y(h - y)/(Bh^3)$

$d\tau/dy = 0 \rightarrow y = h/2$, giving $\tau_{\max} = \mathbf{1.5 \tau_{avg}}$ at mid-height; at the N.A. ($y = 2h/3$) $\tau = (4/3)\tau_{avg}$

Exam tip: Only for sections whose width does not narrow towards the N.A. (rectangle, circle, I) is the peak at the N.A. Triangle and diamond are the classic exceptions.

Q6 Answer: 56.67 (accept 56.1 to 57.2)

Concept: Shear flow along the joint $q = VQ/I$; each nail carries $q \times s$, so $s = F_{\text{nail}}/q$.

Centroid from the bottom: $\bar{y} = (7500 \times 75 + 7500 \times 175)/15\,000 = 125 \text{ mm}$

$$I = 50 \times 150^3/12 + 7500 \times 50^2 + 150 \times 50^3/12 + 7500 \times 50^2 = 5.3125 \times 10^7 \text{ mm}^4$$

$$Q \text{ of the flange about the N.A.} = 7500 \times (175 - 125) = 3.75 \times 10^5 \text{ mm}^3$$

$$q = 3000 \times 3.75 \times 10^5 / 5.3125 \times 10^7 = 21.18 \text{ N/mm} \rightarrow s = 1200/21.18 = \mathbf{56.67 \text{ mm}}$$

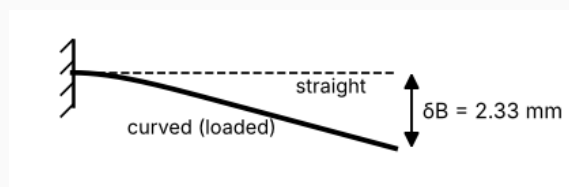
Exam tip: For the joint, Q is the first moment of the part being held on (the flange), not of half the section.

Q7 Answer: (C) Every one of the bolts was checked before assembly.**Concept:** Check subject-verb agreement and tense sequence in each option.(A) With neither...nor, the verb agrees with the nearer subject 'engineers' → *were*.(B) 'each of the samples' is singular → *has failed*.(D) 'Scarcely had ... begun when' needs the past tense → *failed*.(C) 'Every one' is singular and takes 'was' → **correct**.**Exam tip:** Words like each, every one, either and neither (used alone) are singular; the plural noun after 'of' is a trap.**Q8 Answer: (C) extravagant****Concept:** Frugal means economical, careful not to waste money or resources.

Thrifty and prudent are close in meaning (synonyms); modest is unrelated in this sense.

The opposite is **extravagant** (spending wastefully).**Exam tip:** In antonym questions, first eliminate the synonyms that are planted among the options.**Q9 Answer: (A) inconclusive****Concept:** The clause after 'so' gives the consequence; the blank must give a reason to repeat the experiment.

Conclusive and definitive results would end the experiment; 'unanimous' applies to people's opinions, not results.

Only **inconclusive** (not leading to a firm result) explains the repetition.**Exam tip:** Read the connector (so, but, although) first; it tells you whether the blank agrees with or opposes the rest of the sentence.**Q10 Answer: 2.33 (accept 2.31 to 2.35)****Concept:** The loaded 2 m behaves like a cantilever of length $a = 2$ m; beyond it the beam carries no moment and stays straight, inclined at the slope reached at $x = a$.At $x = a$: $\delta_a = wa^4/(8EI) = 6 \times 16/(8 \times 12\,000) = 1.0 \times 10^{-3}$ m; $\theta_a = wa^3/(6EI) = 6 \times 8/(6 \times 12\,000) = 6.667 \times 10^{-4}$ rad $\delta_B = \delta_a + \theta_a(L - a) = 1.000 \text{ mm} + (6.667 \times 10^{-4})(2000 \text{ mm}) = 1.000 + 1.333 = \mathbf{2.33 \text{ mm}}$ **Exam tip:** Forgetting the $\theta_a(L - a)$ term gives 1.00 mm, a common wrong answer.

Q11 Answer: (D) $4PL^3/(243EI)$

Concept: For a load P at distances a and b from the supports, Macaulay's method gives the deflection under the load as $Pa^2b^2/(3EIL)$.

$$a = L/3, b = 2L/3 \rightarrow a^2b^2 = (L^2/9)(4L^2/9) = 4L^4/81$$

$$\delta = P \times 4L^4/(81 \times 3EIL) = 4PL^3/(243EI) \approx 0.0165 PL^3/EI$$

Exam tip: Check: with $a = b = L/2$ the same formula gives $PL^3/(48EI)$. The maximum deflection here lies slightly towards mid-span from the load.

Q12 Answer: 1.88 (accept 1.86 to 1.89)

Concept: For a fixed-fixed beam with a central load, the end moments are $PL/8$ and the deflection is one-quarter of the simply supported value: $\delta = PL^3/(192EI)$.

$$\delta = 40 \times 6^3/(192 \times 24\,000) = 8640/4\,608\,000 = 1.875 \times 10^{-3} \text{ m} = \mathbf{1.88 \text{ mm}}$$

Simply supported, it would be $PL^3/(48EI) = 7.50 \text{ mm}$.

Exam tip: End fixity: moment $PL/4 \rightarrow PL/8$ and deflection $\div 4$. For a UDL the deflection ratio is $1/5$ (Day 10).

Q13 Answer: 0.225 (accept 0.223 to 0.227)

Concept: The length change follows the longitudinal strain $\epsilon_l = (\sigma_l - \nu\sigma_h)/E$.

$$\sigma_h = pd/2t = 1.5 \times 1200/24 = 75 \text{ MPa}; \sigma_l = pd/4t = 37.5 \text{ MPa}$$

$$\epsilon_l = (37.5 - 0.3 \times 75)/200\,000 = 15/200\,000 = 7.5 \times 10^{-5}$$

$$\Delta L = 7.5 \times 10^{-5} \times 3000 = \mathbf{0.225 \text{ mm}}$$

Exam tip: The hoop stress reduces the longitudinal strain through Poisson's effect; ignoring it gives 0.5625 mm .

Q14 Answer: (B) 714.3 N

Concept: Concentric springs of equal length deflect equally, so they act in parallel and share the load in proportion to their stiffness $k = Gd^4/(8D^3n)$.

$$k_o = G \times 10^4/(8 \times 80^3 \times 8) \text{ and } k_i = G \times 6^4/(8 \times 48^3 \times 12) \rightarrow \text{with } G = 80 \text{ GPa, } k_o = 24.41 \text{ N/mm, } k_i = 9.77 \text{ N/mm}$$

$$k_o/k_i = 2.5 \rightarrow W_o = 1000 \times 2.5/3.5 = \mathbf{714.3 \text{ N}}, W_i = 285.7 \text{ N (common deflection 29.3 mm)}$$

Exam tip: G cancels in the ratio, so the answer does not depend on the material, only on $d^4/(D^3n)$.

Q15 Answer: 13.26 (accept 13.1 to 13.4)

Concept: For a gradually applied torque, $U = \frac{1}{2}T\theta = T^2L/(2GJ)$.

$$J = \pi \times 60^4/32 = 1.2723 \times 10^6 \text{ mm}^4$$

$$\theta = TL/(GJ) = 1.5 \times 10^6 \times 1200/(80\,000 \times 1.2723 \times 10^6) = 0.01768 \text{ rad}$$

$$U = \frac{1}{2} \times 1500 \text{ N}\cdot\text{m} \times 0.01768 = \mathbf{13.26 \text{ J}}$$

Exam tip: Equivalent form: $U = (\tau_{\max}^2/4G) \times \text{volume}$ for a solid shaft; check: $\tau_{\max} = 35.37 \text{ MPa}$ gives the same 13.26 J .

What next?

Day 12 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124



Website test series



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Next, Day 13/90: Strength of Materials, Columns & Energy Methods. Comment **GATE90** on our reel to get all links.

Want concise revision notes? The GATE ME handwritten short notes of an AIR 16 topper are available in the Dig Career Thrust app.